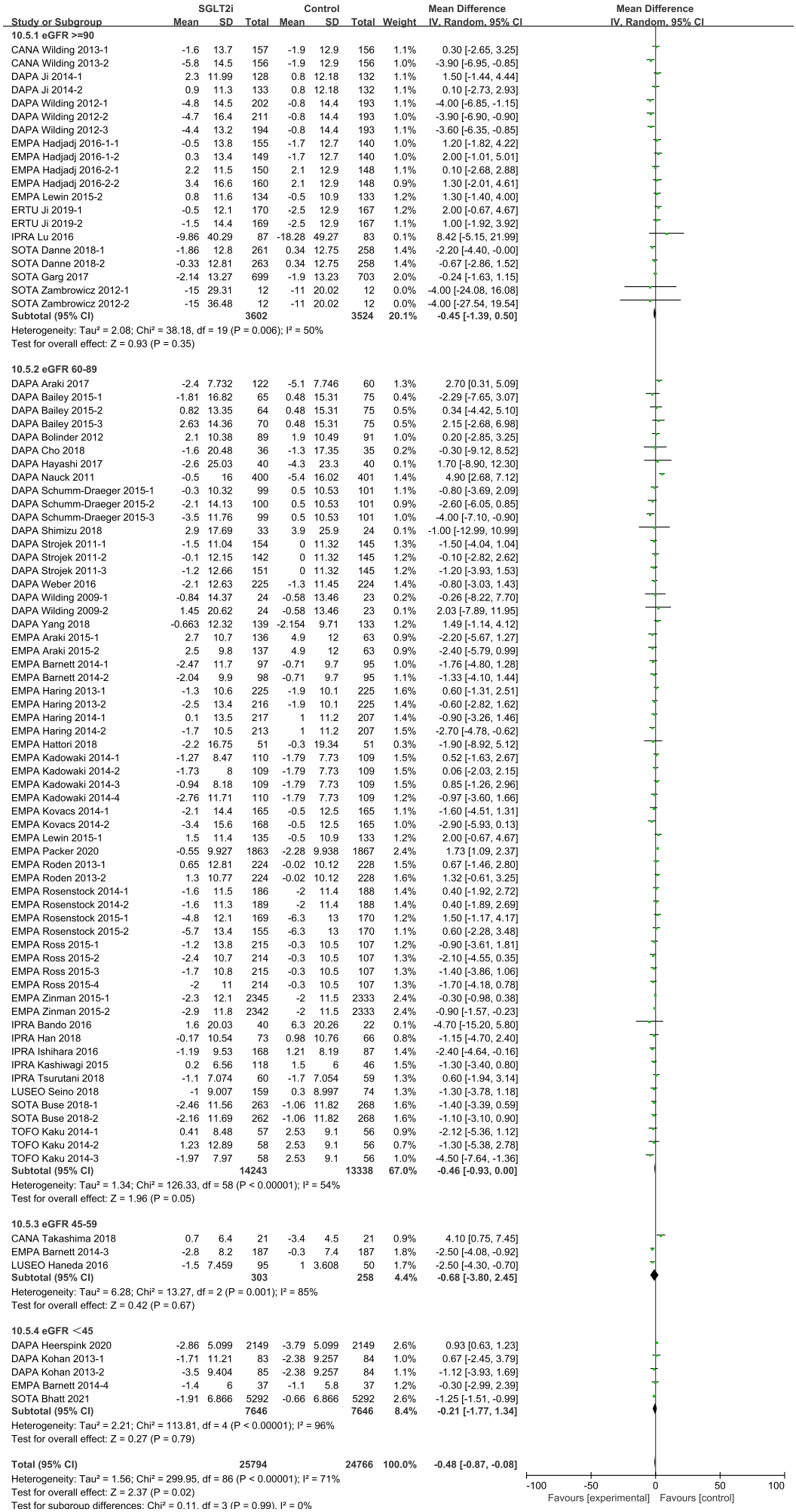

SGLT2i
Control
Mean Difference
Mean Difference
Study or Subgroup Mean SD Total Mean SD Total Weight IV, Random, 95% CI
IV, Random, 95% CI
10.5.1 eGFR >=90
CANA Wilding 2013-1 -1.6 13.7 157 -1.9 12.9 156 1.1% 0.30 [-2.65, 3.25]
CANA Wilding 2013-2 -5.8 14.5 156 -1.9 12.9 156 1.0% -3.90 [-6.95, -0.85]
DAPA Ji 2014-1 2.3 11.99 128 0.8 12.18 132 1.1% 1.50 [-1.44, 4.44]
DAPA Ji 2014-2 0.9 11.3 133 0.8 12.18 132 1.1% 0.10 [-2.73, 2.93]
DAPA Wilding 2012-1 -4.8 14.5 202 -0.8 14.4 193 1.1% -4.00 [-6.85, -1.15]
DAPA Wilding 2012-2 -4.7 16.4 211 -0.8 14.4 193 1.0% -3.90 [-6.90, -0.90]
DAPA Wilding 2012-3 -4.4 13.2 194 -0.8 14.4 193 1.1% -3.60 [-6.35, -0.85]
EMPA Hadjadj 2016-1-1 -0.5 13.8 155 -1.7 12.7 140 1.0% 1.20 [-1.82, 4.22]
EMPA Hadjadj 2016-1-2 0.3 13.4 149 -1.7 12.7 140 1.0% 2.00 [-1.01, 5.01]
EMPA Hadjadj 2016-2-1 2.2 11.5 150 2.1 12.9 148 1.1% 0.10 [-2.68, 2.88]
EMPA Hadjadj 2016-2-2 3.4 16.6 160 2.1 12.9 148 0.9% 1.30 [-2.01, 4.61]
EMPA Lewin 2015-2 0.8 11.6 134 -0.5 10.9 133 1.2% 1.30 [-1.40, 4.00]
ERTU Ji 2019-1 -0.5 12.1 170 -2.5 12.9 167 1.2% 2.00 [-0.67, 4.67]
ERTU Ji 2019-2 -1.5 14.4 169 -2.5 12.9 167 1.1% 1.00 [-1.92, 3.92]
IPRA Lu 2016 -9.86 40.29 87 -18.28 49.27 83 0.1% 8.42 [-5.15, 21.99]
SOTA Danne 2018-1 -1.86 12.8 261 0.34 12.75 258 1.4% -2.20 [-4.40, -0.00]
SOTA Danne 2018-2 -0.33 12.81 263 0.34 12.75 258 1.4% -0.67 [-2.86, 1.52]
SOTA Garg 2017 -2.14 13.27 699 -1.9 13.23 703 2.0% -0.24 [-1.63, 1.15]
SOTA Zambrowicz 2012-1 -15 29.31 12 -11 20.02 12 0.0% -4.00 [-24.08, 16.08]
SOTA Zambrowicz 2012-2 -15 36.48 12 -11 20.02 12 0.0% -4.00 [-27.54, 19.54]
Subtotal (95% CI) 3602 3524 20.1% -0.45 [-1.39, 0.50]
Heterogeneity: Tau² = 2.08; Chi² = 38.18, df = 19 (P = 0.006); I² = 50%
Test for overall effect: Z = 0.93 (P = 0.35)
10.5.2 eGFR 60-89
DAPA Araki 2017 -2.4 7.732 122 -5.1 7.746 60 1.3% 2.70 [0.31, 5.09]
DAPA Bailey 2015-1 -1.81 16.82 65 0.48 15.31 75 0.4% -2.29 [-7.65, 3.07]
DAPA Bailey 2015-2 0.82 13.35 64 0.48 15.31 75 0.5% 0.34 [-4.42, 5.10]
DAPA Bailey 2015-3 2.63 14.36 70 0.48 15.31 75 0.5% 2.15 [-2.68, 6.98]
DAPA Bolinder 2012 2.1 10.38 89 1.9 10.49 91 1.0% 0.20 [-2.85, 3.25]
DAPA Cho 2018 -1.6 20.48 36 -1.3 17.35 35 0.2% -0.30 [-9.12, 8.52]
DAPA Hayashi 2017 -2.6 25.03 40 -4.3 23.3 40 0.1% 1.70 [-8.90, 12.30]
DAPA Nauck 2011 -0.5 16 400 -5.4 16.02 401 1.4% 4.90 [2.68, 7.12]
DAPA Schumm-Draeger 2015-1 -0.3 10.32 99 0.5 10.53 101 1.1% -0.80 [-3.69, 2.09]
DAPA Schumm-Draeger 2015-2 -2.1 14.13 100 0.5 10.53 101 0.9% -2.60 [-6.05, 0.85]
DAPA Schumm-Draeger 2015-3 -3.5 11.76 99 0.5 10.53 101 1.0% -4.00 [-7.10, -0.90]
DAPA Shimizu 2018 2.9 17.69 33 3.9 25.9 24 0.1% -1.00 [-12.99, 10.99]
DAPA Strojek 2011-1 -1.5 11.04 154 0 11.32 145 1.3% -1.50 [-4.04, 1.04]
DAPA Strojek 2011-2 -0.1 12.15 142 0 11.32 145 1.2% -0.10 [-2.82, 2.62]
DAPA Strojek 2011-3 -1.2 12.66 151 0 11.32 145 1.2% -1.20 [-3.93, 1.53]
DAPA Weber 2016 -2.1 12.63 225 -1.3 11.45 224 1.4% -0.80 [-3.03, 1.43]
DAPA Wilding 2009-1 -0.84 14.37 24 -0.58 13.46 23 0.2% -0.26 [-8.22, 7.70]
DAPA Wilding 2009-2 1.45 20.62 24 -0.58 13.46 23 0.1% 2.03 [-7.89, 11.95]
DAPA Yang 2018 -0.663 12.32 139 -2.154 9.71 133 1.2% 1.49 [-1.14, 4.12]
EMPA Araki 2015-1 2.7 10.7 136 4.9 12 63 0.9% -2.20 [-5.67, 1.27]
EMPA Araki 2015-2 2.5 9.8 137 4.9 12 63 0.9% -2.40 [-5.79, 0.99]
EMPA Barnett 2014-1 -2.47 11.7 97 -0.71 9.7 95 1.0% -1.76 [-4.80, 1.28]
EMPA Barnett 2014-2 -2.04 9.9 98 -0.71 9.7 95 1.1% -1.33 [-4.10, 1.44]
EMPA Haring 2013-1 -1.3 10.6 225 -1.9 10.1 225 1.6% 0.60 [-1.31, 2.51]
EMPA Haring 2013-2 -2.5 13.4 216 -1.9 10.1 225 1.4% -0.60 [-2.82, 1.62]
EMPA Haring 2014-1 0.1 13.5 217 1 11.2 207 1.4% -0.90 [-3.26, 1.46]
EMPA Haring 2014-2 -1.7 10.5 213 1 11.2 207 1.5% -2.70 [-4.78, -0.62]
EMPA Hattori 2018 -2.2 16.75 51 -0.3 19.34 51 0.3% -1.90 [-8.92, 5.12]
EMPA Kadowaki 2014-1 -1.27 8.47 110 -1.79 7.73 109 1.5% 0.52 [-1.63, 2.67]
EMPA Kadowaki 2014-2 -1.73 8 109 -1.79 7.73 109 1.5% 0.06 [-2.03, 2.15]
EMPA Kadowaki 2014-3 -0.94 8.18 109 -1.79 7.73 109 1.5% 0.85 [-1.26, 2.96]
EMPA Kadowaki 2014-4 -2.76 11.71 110 -1.79 7.73 109 1.2% -0.97 [-3.60, 1.66]
EMPA Kovacs 2014-1 -2.1 14.4 165 -0.5 12.5 165 1.1% -1.60 [-4.51, 1.31]
EMPA Kovacs 2014-2 -3.4 15.6 168 -0.5 12.5 165 1.0% -2.90 [-5.93, 0.13]
EMPA Lewin 2015-1 1.5 11.4 135 -0.5 10.9 133 1.2% 2.00 [-0.67, 4.67]
EMPA Packer 2020 -0.55 9.927 1863 -2.28 9.938 1867 2.4% 1.73 [1.09, 2.37]
EMPA Roden 2013-1 0.65 12.81 224 -0.02 10.12 228 1.5% 0.67 [-1.46, 2.80]
EMPA Roden 2013-2 1.3 10.77 224 -0.02 10.12 228 1.6% 1.32 [-0.61, 3.25]
EMPA Rosenstock 2014-1 -1.6 11.5 186 -2 11.4 188 1.4% 0.40 [-1.92, 2.72]
EMPA Rosenstock 2014-2 -1.6 11.3 189 -2 11.4 188 1.4% 0.40 [-1.89, 2.69]
EMPA Rosenstock 2015-1 -4.8 12.1 169 -6.3 13 170 1.2% 1.50 [-1.17, 4.17]
EMPA Rosenstock 2015-2 -5.7 13.4 155 -6.3 13 170 1.1% 0.60 [-2.28, 3.48]
EMPA Ross 2015-1 -1.2 13.8 215 -0.3 10.5 107 1.2% -0.90 [-3.61, 1.81]
EMPA Ross 2015-2 -2.4 10.7 214 -0.3 10.5 107 1.3% -2.10 [-4.55, 0.35]
EMPA Ross 2015-3 -1.7 10.8 215 -0.3 10.5 107 1.3% -1.40 [-3.86, 1.06]
EMPA Ross 2015-4 -2 11 214 -0.3 10.5 107 1.3% -1.70 [-4.18, 0.78]
EMPA Zinman 2015-1 -2.3 12.1 2345 -2 11.5 2333 2.4% -0.30 [-0.98, 0.38]
EMPA Zinman 2015-2 -2.9 11.8 2342 -2 11.5 2333 2.4% -0.90 [-1.57, -0.23]
IPRA Bando 2016 1.6 20.03 40 6.3 20.26 22 0.1% -4.70 [-15.20, 5.80]
IPRA Han 2018 -0.17 10.54 73 0.98 10.76 66 0.8% -1.15 [-4.70, 2.40]
IPRA Ishihara 2016 -1.19 9.53 168 1.21 8.19 87 1.4% -2.40 [-4.64, -0.16]
IPRA Kashiwagi 2015 0.2 6.56 118 1.5 6 46 1.5% -1.30 [-3.40, 0.80]
IPRA Tsurutani 2018 -1.1 7.074 60 -1.7 7.054 59 1.3% 0.60 [-1.94, 3.14]
LUSEO Seino 2018 -1 9.007 159 0.3 8.997 74 1.3% -1.30 [-3.78, 1.18]
SOTA Buse 2018-1 -2.46 11.56 263 -1.06 11.82 268 1.6% -1.40 [-3.39, 0.59]
SOTA Buse 2018-2 -2.16 11.69 262 -1.06 11.82 268 1.6% -1.10 [-3.10, 0.90]
TOFO Kaku 2014-1 0.41 8.48 57 2.53 9.1 56 0.9% -2.12 [-5.36, 1.12]
TOFO Kaku 2014-2 1.23 12.89 58 2.53 9.1 56 0.7% -1.30 [-5.38, 2.78]
TOFO Kaku 2014-3 -1.97 7.97 58 2.53 9.1 56 1.0% -4.50 [-7.64, -1.36]
Subtotal (95% CI) 14243 13338 67.0% -0.46 [-0.93, 0.00]
Heterogeneity: Tau² = 1.34; Chi² = 126.33, df = 58 (P < 0.00001); I² = 54%
Test for overall effect: Z = 1.96 (P = 0.05)
10.5.3 eGFR 45-59
CANA Takashima 2018 0.7 6.4 21 -3.4 4.5 21 0.9% 4.10 [0.75, 7.45]
EMPA Barnett 2014-3 -2.8 8.2 187 -0.3 7.4 187 1.8% -2.50 [-4.08, -0.92]
LUSEO Haneda 2016 -1.5 7.459 95 1 3.608 50 1.7% -2.50 [-4.30, -0.70]
Subtotal (95% CI) 303 258 4.4% -0.68 [-3.80, 2.45]
Heterogeneity: Tau² = 6.28; Chi² = 13.27, df = 2 (P = 0.001); I² = 85%
Test for overall effect: Z = 0.42 (P = 0.67)
10.5.4 eGFR <45
DAPA Heerspink 2020 -2.86 5.099 2149 -3.79 5.099 2149 2.6% 0.93 [0.63, 1.23]
DAPA Kohan 2013-1 -1.71 11.21 83 -2.38 9.257 84 1.0% 0.67 [-2.45, 3.79]
DAPA Kohan 2013-2 -3.5 9.404 85 -2.38 9.257 84 1.1% -1.12 [-3.93, 1.69]
EMPA Barnett 2014-4 -1.4 6 37 -1.1 5.8 37 1.2% -0.30 [-2.99, 2.39]
SOTA Bhatt 2021 -1.91 6.866 5292 -0.66 6.866 5292 2.6% -1.25 [-1.51, -0.99]
Subtotal (95% CI) 7646 7646 8.4% -0.21 [-1.77, 1.34]
Heterogeneity: Tau² = 2.21; Chi² = 113.81, df = 4 (P < 0.00001); I² = 96%
Test for overall effect: Z = 0.27 (P = 0.79)
Total (95% CI) 25794 24766 100.0% -0.48 [-0.87, -0.08]
Heterogeneity: Tau² = 1.56; Chi² = 299.95, df = 86 (P < 0.00001); I² = 71%
Test for overall effect: Z = 2.37 (P = 0.02)
Test for subgroup differences: Chi² = 0.11, df = 3 (P = 0.99), I² = 0%
-100 -50 0 50 100
Favours [experimental] Favours [control]